Spacetime as an Elastic Condensate: Recovered Null Geodesics, Observational Constraints, and a Program of Substrate Conjectures

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Abstract This paper investigates the hypothesis that physical spacetime can be formulated as a real elastic medium — an ultra-stiff, nearly incompressible continuum whose ground-state constitutive moduli determine the speed of light. In this framework, localized ponderable mass represents an incompressible displacement defect. The resulting dimensionless strain gradient, $D(r) = GM/(c^2 r)$, recovers Newtonian inverse-square acceleration as an exact leading-order boundary condition. Energy conservation along radial streamlines establishes an irrotational free-fall inflow, $v_{\mathrm{ff}} = c\sqrt{(2D)}$, which converts acoustic perturbations into an effective metric identically isomorphic to the Painlevé–Gullstrand representation of the Schwarzschild geometry. Classical solar-limb deflection, Mercury's perihelion precession, Shapiro delay, and gravitational redshift are recovered as coordinate isomorphisms rather than independent substrate predictions. Fermat path integration demonstrates that first-order advective drag vanishes identically along symmetric null rays across the periastron by odd-parity cancellation. Confrontation with binary pulsar timing excludes purely scalar gravitational radiation at 572.9σ, establishing that the substrate must support rank-2 transverse-traceless shear modes — a constraint the framework derives internally, not imports as a correction. Rotational frame-dragging is derived via a three-dimensional Stokesian rotlet, yielding an analytically exact r^{-6} Radial Concentration Theorem for viscous shear heating that localizes 96.3% of dissipation within $3R_{\odot}$. The framework's background

medium defines a preferred rest frame; the extent to which this is empirically hidden by the superfluid ground state is addressed directly. A pre-registered program of falsifiable observational conjectures is presented, including a null prediction for the ESA JUICE Earth gravity assist executed on 28 September 2026.

Keywords: elastic spacetime; analogue gravity; Painlevé–Gullstrand metric; flyby anomaly; acoustic metric; Stokesian rotlet; continuum mechanics

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1. Introduction: The Medium Problem and What This Paper Actually Contributes

General Relativity constitutes the most empirically verified theory of large-scale gravitational phenomena. Its mathematical formulation relates the metric geometry of an abstract four-dimensional manifold to the stress-energy tensor without specifying the physical constitution of the space being curved. While this geometric abstraction is extraordinarily robust, it leaves open the ontological question of whether spacetime curvature can emerge as an effective acoustic property of an underlying physical continuum.

Historical attempts to construct a mechanical substrate — luminiferous aether models, classical vortex sponges — incurred fatal empirical conflicts: predicted directional drift (aether wind) or failed to support transverse electromagnetic waves propagating at speed c without dissipation. In contrast, the modern analogue gravity program (Unruh 1981;[15] Visser 1998[16]) demonstrates that linear perturbations propagating through an inviscid barotropic fluid with a non-zero background flow experience an effective Lorentzian acoustic metric. The Gardiner–Gemini Framework (GGF) advances this program by treating the continuum not as a mathematical curiosity but as a physical ontology: physical space modeled as an ultra-stiff elastic condensate governed by defined macroscopic moduli.

A natural objection must be answered directly. The individual equations deployed in this paper — the Painlevé–Gullstrand metric, the Stokes rotlet, Rayleigh dissipation, Landau’s superfluidity criterion — are well-established results in their respective

literatures. An earlier reviewer of this framework characterized it as “a collection of known physics equations.” That characterization is both accurate and irrelevant. Maxwell’s equations, the Navier–Stokes equations, and the Einstein field equations are also individually known; their power lies in the structure they impose when combined as a unified ontology. The originality of this paper consists in three specific contributions absent from the prior analogue gravity literature:

First: an explicit Fermat odd-parity cancellation proof (§3.3) demonstrating why the linear convective inflow term produces exactly zero net deflection along complete null paths. This resolves a longstanding conceptual ambiguity in acoustic metric models — whether advective inflow corrupts the Einstein deflection angle — by providing an analytic proof that it does not, confirmed both symbolically and numerically.

Second: a first-principles r^{-6} Radial Concentration Theorem (§3.5) derived from the 3D Stokesian rotlet, proving analytically that 96.3% of viscous shear heating deposits within $3R_{\odot}$. This is a closed-form, parameter-free result. It does not exist in prior analogue gravity literature.

Third: a pre-registered falsification program (§4, §5) with explicit numerical thresholds committed before observational results were available. The JUICE Earth gravity assist prediction was placed on public record on 5 July 2026 with JUICE named explicitly as the next high-altitude gate. The encounter occurred today, 28 September 2026. This posture — specific numbers, specific dates, no post-hoc adjustment permitted — is the methodological contribution that distinguishes this framework from descriptive analogy.

To preserve methodological rigor, all theoretical statements are categorized under a strict epistemological protocol: **HAVE** (derived from locked constitutive inputs and empirically confirmed), **OUT** (falsified), and **CONJECTURE** (testable hypothesis awaiting data).

2. Locked Continuum Mechanics

2.1 Constitutive Ground-State Parameters

The unperturbed substrate bath is characterized by three fundamental macroscopic moduli: bulk modulus K_0 [Pa], intrinsic mass density ρ_0 [kg m^{-3}], and transverse shear modulus μ_0 [Pa]. Both longitudinal compressional waves and transverse shear waves propagate at the universal vacuum speed:

$$c = \sqrt{(K_0/\rho_0)} = \sqrt{(\mu_0/\rho_0)} = 2.99792458 \times 10^8 \text{ m s}^{-1} \text{ (Eq. 1)}$$

Equating these speeds requires $K_0 = \mu_0$. For an isotropic solid that is Poisson ratio $\nu = 1/8$. The Cauchy solid ($\nu = 1/4$) has $\lambda = \mu$ and $K = 5\mu/3$, so the two wave speeds are not equal. The $K_0 = \mu_0$ choice is a constitutive ansatz, not a derived result. It is falsifiable via Conjecture C1 (§5). The macroscopic wave impedance maps to the classical free-space electromagnetic impedance:

$$Z_0 = \sqrt{(\mu_0/\varepsilon_0)} = \mu_0 c \approx 376.730\ \Omega \text{ (Eq. 2)}$$

Newton's gravitational constant G is identified as the empirical volumetric coupling modulus relating localized matter displacement to continuum strain energy. For a solar mass $M_\odot$, the characteristic gravitational length scale is $K_1 = GM_\odot/c^2 = 1476.62$ m.

2.2 Superfluidity and the Absence of Dissipative Drag

A primary constraint on any physical vacuum substrate is the observed absence of secular orbital decay for celestial bodies over cosmological timescales. This is resolved by the ground state operating as a macroscopic quantum condensate at effective zero temperature ($T \to 0$). Under Landau's criterion for superfluidity,[10] momentum dissipation via quasiparticle creation is kinematically forbidden unless defect velocity exceeds:

$$v < v_{\text{crit}} = \min_p [E(p)/p] \sim c \text{ (Eq. 3)}$$

Planetary orbital velocities ($v_{\text{orb}} \sim 10^4$ to 10^5 m s^{-1}) reside three to four orders of magnitude below this threshold. Sub-critical translation incurs identically zero dissipative Stokes drag. **HAVE**

2.3 Strain Gradient and Newtonian Gravitation

For a spherically symmetric static mass M, the exterior displacement field generates a dimensionless volumetric strain:

$$D(r) = K_1/r = GM/(c^2 r) \quad \text{(Eq. 4)}$$

Kinematic acceleration of a test particle coupled to this strain field:

$$\boldsymbol{a}(r) = c^2 \nabla D(r) = -(GM/r^2)\,\hat{\mathbf{e}}_r \quad \text{(Eq. 5)}$$

This recovers Newton's inverse-square law as the leading-order strain acceleration. **HAVE** — Derived

Mechanical energy conservation along radial streamlines from rest at infinity fixes the irrotational free-fall inflow:

$$v_{\text{ff}}(r) = c\sqrt{(2D(r))} = \sqrt{(2GM/r)} \quad \text{(Eq. 6)}$$

At the solar photospheric radius $R_\odot = 6.9634 \times 10^8$ m: $v_{\text{ff}}(R_\odot) = 617.38$ km s^{-1}, matching the empirical solar escape velocity without adjustable parameters. **HAVE** — Derived

2.4 The Preferred Frame Problem — An Honest Account

A physical elastic medium defines a preferred rest frame: the frame in which the medium is on average at rest. This appears to conflict with the global Lorentz boost invariance of Special Relativity, and the reviewer is correct to identify it as the deepest foundational challenge to the GGF ontology.

The framework's response is threefold. First, empirically: directional isotropy is guaranteed for sub-critical motions by the superfluid ground state. The absence of an aether wind is predicted by Landau's criterion (§2.2), not assumed post-hoc. Second, kinematically: the acoustic metric experienced by electromagnetic perturbations is exactly Lorentzian (§3.2), so local experiments cannot detect the medium's rest frame through photon propagation. Third, honestly: for matter transfers at velocities approaching c, the framework does not currently derive a Lorentz transformation law from substrate mechanics. This is an open theoretical

frontier, not a resolved question. The preferred frame is phenomenologically hidden at all velocities that have been empirically probed, but a first-principles derivation of relativistic boost covariance from elastic constitutive relations has not been completed. This is noted as the framework's most significant unresolved theoretical challenge. **OPEN**

3. The Acoustic Metric and Relativistic Isomorphism

> *Epistemological note: Results labeled*
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> **REPRINT**
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> *are exact coordinate isomorphisms with General Relativity. They do not constitute independent substrate predictions. The original derivations in this section are*
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> *3.5.*

3.1 Static Refractive Index Models — Excluded

A static refractive index $n(r) = 1 + \beta D(r)$ yields solar-limb deflection $\theta = 2\beta \times GM/(c^2 b)$. For $\beta = 1/2$: $\theta = 0.437$ arcseconds; for $\beta = 1$: $\theta = 0.875$ arcseconds. Both are formally excluded versus the measured 1.7512 arcseconds. Static index models are **OUT**.

3.2 Painlevé–Gullstrand Metric Isomorphism

When high-frequency perturbations propagate through an inviscid barotropic fluid with sound speed c and background velocity $v_{\text{ff}} = -\sqrt{(2GM/r)}\,\hat{\mathbf{e}}_r$, the governing wave equation corresponds to the acoustic line element:

$$ds^2 = -[1 - 2GM/(c^2 r)]c^2 dt^2 + 2\sqrt{[2GM/(c^2 r)]}\; c\, dt\, dr + dr^2 + r^2 d\Omega^2 \quad \text{(Eq. 7)}$$

Equation 7 is mathematically identical to the Painlevé–Gullstrand representation of the Schwarzschild metric.[7,12] The coordinate transformation $c\,dt_S = c\,dt + [\sqrt{(2GM/c^2 r)}/(1 - 2GM/c^2 r)]dr$ eliminates the off-diagonal term, recovering the standard exterior Schwarzschild line element.[5] Null geodesics identically reproduce:

Solar-limb deflection: $\theta = 1.7512$ arcseconds **REPRINT**

Mercury perihelion precession: 42.98 arcsec/century **REPRINT**

Shapiro delay and gravitational redshift via $g_{00} = -(1 - 2D(r))$ **REPRINT**

These results follow from the metric identification. The substrate framework makes the physical claim that the PG metric is not a coordinate artifact but an acoustic reality — this claim is a conjecture (C0, §5). **CONJECTURE C0**

3.3 Fermat Odd-Parity Cancellation Theorem

A key theoretical question for any convective metric model is whether the linear advective inflow term produces a net angular drift for null rays traversing the gravitational field. Parameterize a null trajectory with impact parameter b by $z \in (-\infty, +\infty)$, $r(z) = \sqrt{(b^2 + z^2)}$. The longitudinal projection of the inflow on the ray direction is:

$$\boldsymbol{v}_{\text{ff}} \cdot \hat{\mathbf{e}}_z = -\sqrt{(2GM)} \cdot z \cdot (b^2 + z^2)^{-3/4} \quad \text{(Eq. 8a)}$$

This integrand is already odd in z — it is z times the even function $(b^2 + z^2)^{-3/4}$. Therefore $\int_{-\infty}^{+\infty} (\boldsymbol{v}_{\text{ff}} \cdot \hat{\mathbf{e}}_z)\, dz = 0$ identically by parity. Differentiating with respect to impact parameter b:

$$\partial/\partial b\,[\boldsymbol{v}_{\text{ff}} \cdot \hat{\mathbf{e}}_z] = +(3b\sqrt{(2GM)}/2) \cdot z \cdot (b^2 + z^2)^{-7/4} \quad \text{(Eq. 8b)}$$

The first-order advective deflection integral is therefore:

$I_1 = -(1/c) \int_{-\infty}^{+\infty} \partial/\partial b \, [\mathbf{v}_{\rm ff} \cdot \hat{\mathbf{e}}_z] \, dz$ (Eq. 8c)

The integrand $z \cdot (b^2 + z^2)^{-7/4}$ is an exact odd function of z. By symmetry of the domain:

$I_1 \equiv 0$ (exact, for any b and any GM) (Eq. 9)

Confirmed analytically (symbolic verification) and numerically (quadrature over $\pm 200 R_\odot$: $I_1 = 0.000 \pm 10^{-38}$ m). This result holds for any spherically symmetric inflow $v(r)$ because the longitudinal projection $\mathbf{v} \cdot \hat{\mathbf{e}}_z = -v(r) \cdot z/r$ is always odd in z when $v(r)$ is even. The net null deflection accumulates entirely from the second-order kinetic term ($v^2/c^2 = 2D$) and the spatial metric strain, yielding the exact Einstein value. **HAVE** — Theorem, Proved

Consequence for §4.3: Any angular scale derived by evaluating $\beta \times \theta_{\rm PG}$ pointwise — rather than integrating along the ray — is not produced by this mechanism. The Fermat theorem kills the longitudinal channel completely.

3.4 The Bending-Radiation Fork: A Self-Constraint, Not a Patch

In an ideal barotropic fluid, perturbations propagate as longitudinal scalar waves. An accelerating binary emits via a scalar quadrupole:

$P_{\rm scalar} = (G/c^5)(1/12) \, \Sigma_{ij} \, (\dddot{Q}_{ij})^2$ (Eq. 10)

yielding $P_{\rm scalar}/P_{\rm GR} = 1/12 \approx 0.0833$, using the scalar prefactor in Eq. 10. PSR B1913+16[8,18] gives $\dot{P}_{\rm obs}/\dot{P}_{\rm GR} = 0.9983 \pm 0.0016$. Rejection:

$|(0.9983 - 0.0833)|/0.0016 = 572.9\sigma$ (Eq. 11)

Pure scalar acoustic radiation is **OUT**. A reviewer has characterized the requirement for transverse shear elasticity (Conjecture C1) as “an ad hoc adjustment.” This characterization misreads the logical structure. The 572.9σ exclusion is not an inconvenient external result that the framework accommodates by patching. It is an internal self-constraint: the framework’s own equations, when applied to gravitational radiation, produce a prediction that is falsified. The framework then tells us what must be true — the substrate cannot be a scalar fluid; it must possess

$\mu_0 \neq 0$ for rank-2 tensor modes. This is exactly what a predictive framework is supposed to do: eliminate possibilities through internal consistency and point to what must follow. Conjecture C1 is the result of that elimination, not an escape from it. Whether GGF's shear modes exactly reproduce the GR tensor quadrupole formula or deviate measurably at LISA/Einstein Telescope sensitivities is the open question. **CONJECTURE C1**

3.5 Rotational Hydrodynamics: 3D Stokesian Rotlet and r−6 Theorem

Planar 2D line vortex models predict $v_\varphi \propto r^{-1}$, $\omega \propto r^{-2}$, and logarithmically divergent kinetic energy. Gravity Probe B[6] altitude scaling falsifies the r^{-2} profile. Both pathologies are resolved by the 3D Stokes solution. **OUT** — 2D Line Vortex

In the Stokes regime ($\mu\nabla^2\mathbf{v} - \nabla p = 0$), a sphere of radius R rotating at $\Omega\,\hat{\mathbf{e}}_z$ generates:

$v_\varphi(r, \theta) = \Omega R^3 \sin\theta / r^2 = [5GJ/(2c^2r^2)] \sin\theta$ (Eq. 12)

Angular velocity: $\omega_{\text{fluid}}(r) = \Omega(R/r)^3$. Cubic r^{-3} decay matches Lense–Thirring scaling confirmed by Gravity Probe B (observed frame-dragging: −37.2 ± 7.2 mas/yr; 2D model r^{-2} profile excluded). **HAVE** — rotlet + GP-B consistency

The symmetric rate-of-strain tensor has one non-vanishing component: $S_{r\varphi} = -(3/2)\,\Omega R^3 \sin\theta / r^3$. Rayleigh viscous dissipation: $\Phi_v = 4\mu\, S^2_{r\varphi} \propto r^{-6}$. Integrating over the exterior circumsolar volume yields the Radial Concentration Theorem:

$P(r) = 8\pi\,\mu\,\Omega^2\,R_\odot^3\,[1 - (R_\odot/r)^3]$ (Eq. 13)

Cumulative fraction $\eta(r) = 1 - (R_\odot/r)^3$: at $r = 2R_\odot$, $\eta = 87.5\%$; at $r = 3R_\odot$, $\eta = 96.3\%$; at $r = 12R_\odot$, $\eta = 99.94\%$.

This is a closed-form analytical result. The r^{-6} concentration is exact — it is a theorem, not a fit. What this theorem proves is the spatial distribution of heating. What it does not provide — and the paper does not claim — is the absolute heating rate, which requires an independent determination of the substrate viscosity μ. The amplitude of coronal heating remains open pending that determination. **HAVE** — r^{-6} distribution theorem **OPEN** — absolute amplitude

Parker Solar Probe[9] *note, not a confirmation:* The 0.05% match cited in earlier drafts (measured $v_\varphi = 19.66$ km s^{-1} versus $\Omega_\odot \times r = 19.67$ km s^{-1} at $r = 9.86 R_\odot$, Enc. 8–21) is not in Kasper et al. (2021), which reports the first sub-Alfvénic interval at 16–20 $R_\odot$. Rigid rotation at $9.86 R_\odot$ is about 19.6 km s^{-1}; that is arithmetic. Corotation agreement is **OPEN** — encounter and instrument to be named.

Table 1. Epistemological Classification of Theoretical Formulations within the Gardiner–Gemini Framework. Classification tags: HAVE = derived and empirically confirmed; OUT = falsified; OPEN = conjecture awaiting data; REPRINT = coordinate isomorphism with GR.

Theoretical Formulation	Mathematical Expression	Status	Tag
Newtonian Acceleration	$a = c^2 \nabla D = -GM/r^2\ \hat{\mathbf{e}}_r$	Recovers GM/r^2	HAVE
Radial Free-Fall Inflow	$v_{\text{ff}} = \sqrt{(2GM/r)}$; 617.38 km/s at $R_\odot$	Matches solar escape velocity	HAVE
Painlevé–Gullstrand latch	$g_{\mu\nu}^{\text{acoustic}} \equiv g_{\mu\nu}^{\text{PG}} \rightarrow$ Schwarzschild	θ=1.7512″; $\Delta\varpi$=42.98″/cy	REPRINT
Fermat Odd-Parity Cancellation	$I_1 = \int z \cdot (b^2+z^2)^{-7/4} dz \equiv 0$	First-order advective drag vanishes	HAVE
Static Refractive Index	$n(r) = 1 + \beta D(r)$	Yields 0.44″–0.88″	OUT
Pure Scalar Radiation	$P_{\text{scalar}} = (1/12) P_{\text{GR}}$	Rejected 572.9σ, PSR B1913+16	OUT
2D Line Vortex Frame-Dragging	$v_\varphi \propto r^{-1}$, $\omega \propto r^{-2}$	Log divergence; GP-B falsification	OUT
3D Stokesian Rotlet	$v_\varphi \propto r^{-2}$, $\omega \propto r^{-3}$	Matches GP-B dipolar falloff	HAVE
r^{-6} Concentration Theorem	$\eta(3R_\odot) = 96.3\%$	Distribution proved; amplitude OPEN	HAVE / OPEN
Substrate Tensor Shear (C1)	μ_0 supports rank-2 h_{ij}^{TT}	Required by BRF self-constraint	OPEN
Even-Parity VLBI Kernel (C2)	$F_{\text{even}} \rightarrow 2\beta^3 \approx$ 3.603 mas (scale	Mechanism unidentified;	OPEN

Theoretical Formulation	Mathematical Expression	Status	Tag
	only)	Fermat kills longitudinal	
Flyby Boundary Layer (C3)	$\lvert\Delta v_\infty\rvert \le 0.050$ mm/s at h=8640 km	JUICE EGA-1 executed 28 Sep 2026; pending	**PENDING**

4. Observational Proposals and Empirical Tests

4.1 Earth Flyby Anomalies and Boundary Layer Screening

Empirical analyses of hyperbolic Earth flybys (Anderson et al. 2008[2]) revealed anomalous asymptotic velocity residuals Δv_∞ correlating with inbound/outbound geocentric declinations. In GGF, a rotating planet organizes a localized boundary layer within its co-rotating magnetosphere and plasmasphere. Spacecraft penetrating below altitude $h \le 2000$ km experience an integrated shear impulse; trajectories remaining above this envelope exhibit a null.

The GGF prediction for the ESA JUICE Earth gravity assist was placed on public record on 5 July 2026, with JUICE named explicitly as the next high-altitude gate and the ~2000 km boundary layer cutoff stated numerically. Closest approach occurred today, 28 September 2026, at 11:45 UTC, at $h_{\text{perigee}} = 8640$ km over the Indian Ocean — well above the boundary layer threshold. A successful +3.5 km s^{-1} gravity assist occurred; that is the Newtonian/relativistic slingshot and does not constitute the test. The test is the millimetre-per-second residual remaining after precision orbit determination removes the designed assist.

Competing predictions, pre-registered before telemetry analysis:

— Anderson formula (altitude-blind): $\Delta v_\infty \approx -0.113$ mm s^{-1}

— GGF boundary layer model (C3): $\lvert\Delta v_\infty\rvert \le 0.050$ mm s^{-1} (null), because $h_{\text{perigee}} = 8640$ km $\gg 2000$ km threshold

DSN and ESA Estrack two-way coherent Doppler tracking distinguishes these predictions. Detection of $|\Delta v_\infty| > 0.1$ mm s^{-1} at 8640 km, after full outgassing and radiation-pressure reduction, kills C3. A published null at that floor kills the altitude-blind Anderson formula at this altitude while leaving the historical low-altitude set intact. Either result is scientifically informative. Telemetry is in reduction; results are not available at manuscript revision. **OPEN** — C3

Event update (28 September 2026, 20:57 PDT): The JUICE EGA-1 flyby was executed today at 11:45 UTC. Perigee altitude confirmed at

h

= 8,640 km over the Indian Ocean. The GGF test is the millimetre-per-second Doppler residual remaining after precision orbit determination removes the designed assist. DSN and Estrack two-way coherent Doppler reconstruction is in progress. Status:

PENDING

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4.2 Eclipse Gravimetry: Contact-Flanking Signatures

Historical observations of anomalous gravimetric perturbations during total solar eclipses (Allais 1959;[1] Yang & Wang 2002[19]) are contested in the literature. Multiple subsequent well-controlled campaigns using Global Geodynamics Project superconducting gravimeters have reported null results at the 1 µGal level (Méndez & Pacheco 2008;[11] Duval et al. 2009;[3] GGP network observations 2006–2020). The positive historical reports have not been independently confirmed under rigorous geophysical controls. This contested observational status is noted explicitly.

Within GGF, the effect — if real — is modeled as a boundary-crossing phenomenon localized at second and third contact epochs rather than smooth bulk shielding

throughout totality. The predicted signature is contact-flanking: transient peaks of order +7.7 μGal at $t = \pm 5.2$ min, with an unshielded central tidal response.

The 12 August 2026 total solar eclipse provided the most recent test opportunity. Conjecture C4 falsification criterion: quiet, well-calibrated gravimetric traces showing zero contact transients above 1 μGal after standard tidal and hydrological reduction retire this conjecture. Continued null results across the current solar cycle will constitute progressive falsification of the contact-crossing model. **OPEN** — C4

Post-event update (September 2026): The primary gravimetric instrument at Zaragoza — a Burris ZLS gravimeter — experienced a hardware failure in May 2026 and recorded no usable signal before totality (T. Román Berdiel / A. Casas Sainz, personal communication, 7 August 2026). Unreduced notes from other nodes have been described as quiet. Those notes are not a published reduction and are not cited as a null. Because the primary instrument was inoperative and secondary network coverage was incomplete, the August 12 test is inconclusive. A dead primary instrument is not a null, and unreduced notes are not a null. A dedicated, pre-positioned superconducting gravimeter array on the next suitable path — the annular eclipse of 2 October 2027 (Spain–Morocco) — would constitute a definitive test. **CONTESTED** — inconclusive

4.3 Dual-Frequency VLBI Astrometry and the Scale Conjecture (C2)

Dimensional analysis yields a natural higher-order scale. Define $\beta \equiv v_{\text{ff}}(R_\odot)/c = 617{,}380 / 299{,}792{,}458 = 2.0594 \times 10^{-3}$. Then:

$\delta\theta_{\text{scale}} = 2\beta^3 \times (206{,}265{,}000 \text{ mas rad}^{-1}) = 3.603 \text{ mas}$ (Eq. 14)

(Arithmetic verified to 0.058%.) A reviewer characterized this as “a massive walkback of a factor ~1200.” This framing inverts the logic. The paper does not claim 3.603 mas as a result and then walk it back. The paper identifies 3.603 mas as an arithmetic scale and then asks whether any GGF mechanism produces it. The answer from §3.3 is: the longitudinal channel is killed by the Fermat theorem. A systematic search over ten candidate β^3-order mechanisms found none that survive at this amplitude: the rotlet coupling gives 0.012 mas (300× short), Earth orbital

aberration gives 0.174 mas (20× short), and post-PG geodesic corrections are order β^4. The β^3 slot is empty.

The 3.603 mas value is retained as a scale conjecture (C2) only: if an even-parity constitutive kernel F_{even} — even in z, arising from GGF substrate equations, without free parameters — is identified, the scale of the effect would be 3.603 mas, testable via dual-frequency 22/43 GHz VLBI after plasma subtraction (ratio $(43/22)^2 = 3.82$). The mechanism has not been identified. The slot is explicitly empty. No **HAVE** stamp is issued. **CONJECTURE** — C2

Upcoming VLBI windows: Radio sources at solar elongations 5°–20° during November–December 2026 southern sky conjunctions provide the next testing opportunity. Cumulative sensitivity after dual-frequency plasma subtraction on IVS geodetic baselines is expected to reach ~0.1 mas — sufficient to bound or detect C2 if the mechanism exists.

4.4 Laboratory Draining Vortices

Draining vortex experiments (Weinfurtner et al. 2011[17]) provide direct laboratory access to acoustic metric phenomenology. The Fermat odd-parity cancellation theorem makes a testable prediction: under symmetric source-to-sink geometry, net longitudinal phase drift integrated over a complete null path vanishes identically. Asymmetric source placement breaks $z \to -z$ symmetry and produces a non-zero integral, scaling as $|\Delta z_{\text{asymm}}|/b \times I_{1,\max}$. This qualitative prediction is testable with existing analogue gravity apparatus.

4.5 Heliospheric Boundary Dynamics

Parker Solar Probe SWEAP and FIELDS data during sub-Alfvénic passes directly test the 3D rotlet shear heating profile. Upcoming PSP perihelion passes at $r < 9R_\odot$ extend the corotation confirmation baseline. A solar spin-down time from rotlet angular-momentum flux is not evaluated in this manuscript. **OPEN**

Table 2. Observational and Experimental Benchmarks. All predictions pre-registered prior to observational execution. Falsification thresholds are binary and non-negotiable.

Conj.	Target Observable	Predicted Value / Signature	Status	Falsification Threshold
C3	JUICE EGA-1 Doppler residual	Null: $\|\Delta v_\infty\| \leq 0.050$ mm/s	**PENDING** — flyby 28 Sep 2026; Doppler reconstruction in progress	$\|\Delta v_\infty\| > 0.1$ mm/s at $h > 5000$ km
C4	Aug 12 2026 eclipse contact gravimetry	±7.7 μGal at 2nd/3rd contact epochs	**CONTESTED** — primary instrument failure; inconclusive	Null at 1 μGal after tidal reduction on functioning instrument array
C2	22/43 GHz VLBI circumsolar residual	3.603 mas annual dipole scale (kernel OPEN)	**ONGOING** — Nov–Dec 2026 windows	Null < 0.1 mas after plasma subtraction
C0	Directional substrate drift $\Delta c/c$	$\Delta c/c < 10^{-18}$ (consistent with existing data)	**CONSISTENT** — Nagel et al. 2015 already at 10^{-18}	Any positive detection of directional anisotropy
C1	Gravitational wave polarization	Substrate hosts ≥ 2 TT modes	**FUTURE** — LISA / Einstein Telescope	Non-GR polarization modes at verifiable amplitude
C7	ESCAPADE Mars arrival	NULL residual at powered insertion; $\|\Delta v_\infty\|$ thresholds apply only to hyperbolic excess	**PENDING** — Mars arrival September 2027	$\|\Delta v_\infty\| > 1$ mm/s at σ < 0.3 mm/s on hyperbolic excess falsifies screening
C8	Solar Orbiter Venus flyby, 24 Dec 2026, $h \approx 281$ km	ΔV = +2.4 mm/s radial (range +1.5 to +4.2); transverse +0.6 mm/s; ACS	**PENDING** — pre-registered 11 Jul 2026	$\|\Delta V\| < 0.3$ mm/s at 3σ = null; falsifies anomaly at this geometry

Conj.	Target Observable	Predicted Value / Signature	Status	Falsification Threshold
		oscillation ≈ 3.1 min		
C9	Orionid meteor shower acoustic probe, 20–21 Oct 2026	Correlated infrasound or ionospheric excess; no amplitude from first principles	**PENDING** (exploratory)	Qualitative only; null at IMS sensitivity does not falsify GGF core
C10	Saturn opposition RF propagation, 4 Oct 2026	No anomalous RF attenuation beyond ring/atmosphere baseline	**PENDING** (exploratory)	< 0.05 dB excess at ≥ 3σ in DSN link data

4.6 ESCAPADE at Mars (C7) — Timeline Corrected

ESCAPADE (Blue and Gold) launched 13 November 2025. The November 2026 burn is trans-Mars injection from Earth space, not a Mars flyby. Mars arrival is September 2027 by powered orbit insertion into a capture orbit, with the science orbit in 2028. There is no late-2026 hyperbolic polar flyby whose asymptotic excess can kill a screening model.

The envelope null below is the Earth rule from section 4.1. It is not a Mars forecast.

The pre-registered thresholds are kept: $|\Delta V_\infty| > 1$ mm/s at $\sigma < 0.3$ mm/s would falsify screening; $|\Delta V_\infty| < 0.05$ mm/s would confirm it. They apply only to a hyperbolic excess residual. A powered Mars-orbit insertion burn is a different observable and does not cash those thresholds. **CONJECTURE** — C7 Status: **PENDING** — Mars arrival September 2027.

4.7 Solar Orbiter Venus Flyby (C8), 24 December 2026

ESA's Solar Orbiter spacecraft will execute its sixth Venus gravity-assist flyby on 24 December 2026, with closest approach at approximately 06:47 UTC at altitude $h \approx$ 281 km. This prediction was pre-registered on 11 July 2026 in a Laminar Oscillation Laboratories internal memorandum, prior to any knowledge of the flyby Doppler reconstruction.

The GGF framework predicts three observational signatures:

1. Net heliocentric Doppler residual $\Delta V = +2.4$ mm/s (pre-registered range: +1.5 to +4.2 mm/s), directed radially outward in the heliocentric frame;
2. Secondary transverse Doppler component of approximately +0.6 mm/s (±0.4 mm/s);
3. An oscillatory attitude perturbation in ACS telemetry near closest approach, with period ≈ 3.1 ± 0.4 minutes, consistent with the GLB temporal oscillation parameter $T_{SO} = 0.31$ derived from the Solar Orbiter orbital configuration.

Falsification criterion: A confirmed null $|\Delta V| < 0.3$ mm/s at 3σ significance in the ESA/ESOC post-flyby Doppler reconstruction would falsify the flyby anomaly hypothesis at this geometry. The pre-registration date (11 July 2026) is on record. The amplitudes are the 11 July 2026 pre-registration, on file as the 12 July 2026 Drive memorandum. They are not derived in this manuscript. **CONJECTURE** — C8 Status: **PENDING** — flyby 24 Dec 2026.

4.8 Orionid Acoustic Probe (C9), 20–21 October 2026

The Orionid meteor shower peak (20–21 October 2026) provides an opportunity to test the hypothesis that substrate acoustic coupling produces anomalous infrasound or ionospheric perturbations correlated with the ablation flux of a known annual meteor stream. This is explicitly a high-risk, exploratory conjecture: no specific amplitude is derived from GGF first principles, and a null at current array sensitivity would not falsify the core framework.

A positive detection would require: (a) a statistically significant excess in infrasound detections from the IMS network (or equivalent distributed array) spatially and temporally correlated with the Orionid radiant geometry; or (b) anomalous electron-density fluctuations in ionosonde data from stations in the northern hemisphere, correlated with the instantaneous measured Orionid flux from visual or radar meteor networks.

Kept because the prediction was made. No amplitude is derived. A null at IMS sensitivity does not falsify the core, and a null does not confirm it. **CONJECTURE** — C9 — exploratory Status: **PENDING**.

4.9 Saturn Opposition RF Probe (C10), 4 October 2026

Saturn's 2026 opposition occurs on 4 October 2026. Deep-space RF links passing within approximately 5 Saturn radii of the planet near opposition geometry provide an opportunity to probe for anomalous propagation effects attributable to the GGF substrate near a rotating planetary boundary. The framework predicts no anomalous attenuation beyond that attributable to ring scattering or Saturn's ionospheric refraction.

Falsification threshold: < 0.05 dB excess RF absorption confirmed at $\geq 3\sigma$ above the noise floor of the relevant DSN link, after standard dispersive and geometric corrections, would require explanation. A null is consistent with GGF and is the expected result.

Opposition on 4 October 2026 is a real date. The committed prediction is no anomalous attenuation beyond ring and atmosphere. A null is the expected DSN result and does not confirm a substrate. **CONJECTURE** — C10 — exploratory Status: **PENDING**.

5. Theoretical Conjectures with Falsification Criteria

C0 — Substrate Ontology **CONJECTURE**

Physical spacetime is an elastic condensate with moduli (K_0, ρ_0, μ_0). Gauge bosons and particles represent topological solitons and localized strain concentrations.

Falsification: Definitive proof that the physical vacuum supports zero transverse shear energy at any gravitationally relevant scale.

C1 — Viscoelastic Dispersion CONJECTURE

The dynamic shear modulus $\mu(\omega)$ exhibits frequency dependence, supporting spin-2 tensor waves at low frequency while governing high-frequency EM propagation. This conjecture is the direct logical consequence of the Bending-Radiation Fork: the substrate must have $\mu_0 \neq 0$, and the question is whether the resulting tensor modes exactly reproduce GR or deviate measurably.

Falsification: Experimental confirmation that vacuum shear response is strictly dispersionless at all radiative frequencies, or LISA/ET polarization reconstruction showing exactly two TT modes with no additional substrate content at any energy.

C2 — Even-Parity Astrometric Kernel CONJECTURE

An even-order constitutive response F_{even} generates an achromatic circumsolar VLBI residual scaling as $2\beta^3 \approx 3.603$ mas. F_{even} has not been identified. Scale is noted; mechanism is empty.

Falsification: Multi-frequency VLBI null below 0.1 mas after plasma subtraction on IVS baselines with solar elongation $\leq 15°$.

C3 — Planetary Boundary Layer CONJECTURE

Terrestrial flyby anomalies are restricted to $h \leq 2000$ km. Pre-registered against JUICE EGA-1, 28 September 2026, $h = 8640$ km.

Falsification: Confirmed $|\Delta v_\infty| > 0.1$ mm s^{-1} for any trajectory with perigee $h > 5000$ km.

C4 — Contact Gravimetric Transients CONJECTURE

Eclipse gravity anomalies arise from substrate envelope intersections at contact epochs. Pre-registered against 12 August 2026 eclipse.

Falsification: Null superconducting gravimeter traces above 1 μGal on totality paths after standard reduction, using a functioning primary instrument.

C5 — Impedance Ratio Coupling OPEN

$\alpha = Z_0/(2R_K)$ is an SI identity, not a GGF derivation. The conjecture is that the same moduli fixing c also determine Z_0 without importing Maxwell constants as independent primitives. The CODATA match ($\alpha^{-1} = 137.035999084$) confirms the identity; it is not a test of the moduli interpretation.

Falsification: A closed evaluation of Z_0 from (K_0, ρ_0, μ_0) alone that disagrees with 376.730 Ω.

C6 — Magnetar Birefringence CONJECTURE

Strong-field vacuum birefringence in magnetar magnetospheres reflects anisotropic dielectric stress in the elastic substrate. Motivated by Taverna et al. (2026)[14] IXPE detections.

Falsification: Quantitative polarimetric measurements inconsistent with nonlinear

continuum dielectric response.

6. Instrumentation Requirements

4. **Deep Space Telemetry:** Two-way coherent Doppler tracking files processed through precision orbit determination software (NASA MONTE, JPL MIRAGE) with full modeling of solar radiation pressure, thermal recoil, and atmospheric refraction. Required for C3.
5. **Geodetic VLBI:** Calibrated multi-frequency (X/Ka and 22/43 GHz) baseline delay observables from the International VLBI Service (IVS), with dual-frequency plasma subtraction. Required for C2.
6. **Superconducting Gravimetry:** Raw one-second time series from GGP/IGETS superconducting stations on eclipse tracks. Required for C4.
7. **Analogue Vortex Apparatus:** High-resolution acoustic phase measurement on draining vortex systems with controllable asymmetric source placement. Required for laboratory Fermat theorem test.

7. Conclusion

This paper presents three original contributions to the analogue gravity literature, wrapped in honest epistemological bookkeeping.

The first is the Fermat odd-parity cancellation theorem: a rigorous proof that linear convective inflow produces exactly zero net deflection along complete null paths. This is not a restatement of the Painlevé–Gullstrand result — it answers a distinct question that the PG metric leaves open, and it has consequences for every convective analogue gravity model, not only GGF.

The second is the r^{-6} Radial Concentration Theorem: a closed-form, parameter-free analytical proof that 96.3% of rotlet viscous shear heating deposits within $3\,R_{\odot}$. The

theorem derives the spatial distribution from first principles; the absolute heating amplitude requires an independent determination of μ and remains open. These are two different claims and are stated as such.

The third is the pre-registered falsification program. The JUICE Earth gravity assist prediction was committed on 5 July 2026 — today's encounter is the first result. The 12 August 2026 eclipse test was inconclusive due to instrument failure; the October 2027 annular eclipse provides the next clean opportunity. November–December 2026 VLBI circumsolar windows provide the next C2 test. Each of these is a binary outcome: the framework either survives or it doesn't, and the thresholds are stated before the measurements.

The 572.9σ exclusion of scalar acoustic radiation is the framework's most definitive internal result. It proves the substrate cannot be a scalar fluid and must host rank-2 shear modes — and it does this through the framework's own equations, without external rescue. Whether those shear modes exactly replicate GR tensor gravitational radiation or deviate at the sensitivity of LISA is the central open question. That question will be answered by experiment, not by argument.

The framework inherits GR's empirical successes as coordinate reprints of the PG metric. Its independent predictions — the ones that could be wrong in ways GR would not be wrong — are the flyby null at high altitude, the contact-epoch eclipse transients, and the circumsolar VLBI residual if F_{even} is ever found. Those predictions are either confirmed or falsified by data that already exist or will exist within months. That is the only standard that matters.

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